

Econometrics Stata Lab:

Does School Spending Raise Test Scores?

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Week 5, Fall 2026

Running example: Papke (2005), "The effects of spending on test pass rates: evidence from Michigan."

Builds on Week 4: pooled OLS, fixed effects, and dummy variables.

1. **Recap of Week 4:** compare a school with itself; dummies as intercepts.
2. **The policy:** Michigan's 1994 funding reform (Proposal A).
3. **Real schools:** everything rose together $\Rightarrow$ year dummies.
4. **Papke's regression:** pooled OLS vs. school fixed effects.
5. **Reading the table:** standard errors, stars, and what fixed effects cannot fix.

Recap: Week 4 in one slide

- **Pooled OLS** puts one line through all school-years. With Maple, Oak and Pine it gave the wrong sign (-36.7).
- **Fixed effects** compares each school with itself: subtract each school's average, or run `xtreg`, `fe`. Slope 10.
- **School dummies** give each school its own intercept (Maple 38, Oak 51, Pine 69, at log spending 8) with one shared slope. Same answer: 10.

Today: the same tools on 1,773 real Michigan schools, plus **year** dummies.

Recap: the rest of last week's xtreg output

Line	What it means
Std. err. ≈ 0 , R^2 within = 1	The data are made up and lie exactly on a line. Real data will never look like this.
$\text{sigma_u} = 15.57$	How spread out the school intercepts are (-42 , -29 , -11). Schools differ a lot in their usual level.
$\text{sigma_e} \approx 0$	Left-over noise within a school. It's zero only because the data are made up.
$\rho = 1$	The share of the unexplained variation that comes from differences between schools. Here it's all of it.
$\text{corr}(u_i, Xb) = -0.86$	The most important line. Schools with higher spending have lower intercepts. Maple spends the most and scores the lowest: exactly why pooled OLS gets the sign wrong.
F test that all $u_i = 0$	Tests whether the school intercepts are all equal. They clearly aren't, so pooled OLS (one intercept for everyone) doesn't fit this data.
R^2 between / overall	You can skip these for now.

Begin with two questions

Before any regression, check the data:

1. **What does one row represent?**

One Michigan school in one school year (year = ending year, so 1993 means 1992/93).

2. **Which variables uniquely identify a row?**

The key is `schid year`.

One row is one school in one year. The school ID by itself is not enough, because the same school appears in several years.

In Stata, we used:

- `isid schid year`

The paper's question

Papke (2005) asks: **when a Michigan school gets more spending per pupil, do more fourth-graders pass the math test?**

- The outcome is the school's fourth-grade math pass rate (`math4`).
- Spending is averaged over this year and the previous year, then logged (`lavgrexp`).
- Each dot is a school in a school year. It is not an individual student.

First ask what two schools have in common and how they differ. Then ask what happens to one school as it changes over time.

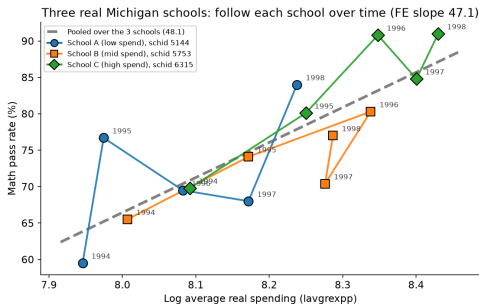
Why do Michigan schools change over time?

In 1994, Michigan changed school funding through Proposal A.

- Before Proposal A, districts relied heavily on local funding and spending differences had grown.
- The reform raised a minimum funding level and limited spending growth for high-spending districts.
- The changes gave Papke school-by-year data with spending that moved over time, not just a snapshot of different schools.

This policy setting makes it useful to follow a school over time. Later, Papke uses the grant rule as another way to separate spending changes from other changes that might affect test scores.

Three real Michigan schools: everything rose together



Schools 5144, 5753, 6315 (1994–1998). Every path climbs steeply (slope 47!). But that is not only money: from 1994 to 1998 spending *and* pass rates rose in almost every Michigan school (average pass rate 51% → 74%, partly because the test changed). Following a school over time also follows the calendar.

All schools, fixed effects: slope ≈ 45 without year controls, ≈ 7 with them. That is why Papke adds year effects (`i.year`).

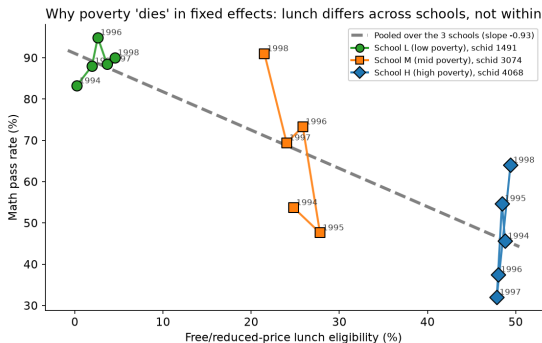
Year dummies: one intercept per year

Last week: one dummy per **school** gave each school its own intercept.
Now: one dummy per **year** gives each year its own intercept.

$$\text{math4}_{it} = \alpha_i + \gamma_{1995} d_t^{1995} + \cdots + \gamma_{1998} d_t^{1998} + \beta \text{lavgrexpp}_{it} + \cdots$$

- $d_t^{1998} = 1$ in 1998 and 0 otherwise. i.e. year makes these dummies.
- γ_{1998} = how much higher every school scores in 1998 than in the base year 1994, whatever it spent (e.g. the test changed).
- The base year is left out, as Maple was: the dummy variable trap again.
- Without year dummies, that statewide rise is credited to spending (slope ≈ 45). With them, ≈ 7 .

Why free-lunch status changes little inside a school



lunch differs enormously *across* schools but barely moves *within* a school over five years (within $SD \approx 1.6$ pp vs between $SD \approx 23$ pp here). The same-school comparison has little movement in lunch to use. In the paper, its coefficient moves from -0.58 in pooled OLS to -0.15 with school effects and is no longer statistically significant.

What Papke found

	Pooled	School FE
Published coefficient on <code>lavgrexpp</code>	8.442	7.179
Approx. pass-rate change per 10% more spending	+0.84 pp	+0.72 pp
Observations in article	7,242	7,242

Both estimates are positive. Papke's fixed-effects estimate is smaller, but remains statistically significant. He later uses the foundation-grant rule to instrument for spending; his conclusion summarizes the effects as roughly 1–2 percentage points per 10% more spending, with larger effects at initially low-performing schools.

The article divides the log-spending coefficient by 10 as a rule of thumb.

Our Stata table: pooled vs. fixed effects

	Pooled OLS	Fixed effects
Log spending (lavgrexpp)	7.638*** (1.721)	6.270** (2.432)
Free lunch (%)	-0.579*** (0.041)	-0.156 (0.096)
Free lunch squared	0.002*** (0.001)	0.001 (0.001)
Log enrollment	-3.992 (8.086)	-12.262 (15.687)
Log enrollment squared	0.207 (0.710)	0.896 (1.367)
Year 1995	12.189*** (0.550)	11.672*** (0.534)
Year 1996	13.258*** (0.647)	13.092*** (0.689)
Year 1997	10.458*** (0.670)	10.152*** (0.733)
Year 1998	23.389*** (0.697)	23.464*** (0.765)
Observations	7,274	7,274
Schools		1,773

Standard errors in brackets, clustered by school. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Base year 1994.

Spending (lavgrexpp): $7.6 \rightarrow 6.3$. Both positive; FE a bit smaller. 10% more spending $\approx +0.6$ to $+0.8$ points.

Poverty (lunch): $-0.58 \rightarrow -0.16$, no longer significant. Poverty differs a lot *between* schools but barely moves *inside* a school, so FE has little to learn from.

Year effects: 1998 is about 23 points above 1994 in *both* columns, for every school. Without `i.year`, that statewide jump would be credited to spending.

What fixed effects handles, and what it does not

It handles

School characteristics that stay the same over these years, even if we did not measure them: for example, a school's long-standing location or culture.

It does not automatically handle

Changes over time that affect both spending and scores, such as a changing group of students or a new school policy.

So fixed effects makes the comparison more credible, but it does not by itself prove that spending caused the score change. Papke also studies the funding reform as a source of spending variation.

Run it in Stata

The Week 5 do-file `week05_papke.do` runs (change the `cd` line first):

```
bcuse school93_98, clear
xtset schid year
gen lunch2 = lunch^2
gen lenrol2 = lenrol^2
regress math4 lavgrexpp lunch lunch2 lenrol lenrol2 i.year, vce(cluster schid)
xtreg math4 lavgrexpp lunch lunch2 lenrol lenrol2 i.year, fe vce(cluster schid)
```

`xtset` tells Stata which school and year identify the observations. `xtreg`, `fe` runs the same-school comparison. Optional Appendix A plots the real schools and the free-lunch confidence intervals.

Leslie E. Papke (2005), “The effects of spending on test pass rates: evidence from Michigan,” *Journal of Public Economics* 89, 821–839.

Table 7, columns 1–2: pooled OLS and school fixed effects of math4 on lavgrexpp, lunch, lunch², lenrol, lenrol², and year effects.

Teaching examples: three selected Michigan schools (schids 5144, 5753, 6315) and three by poverty level (1491, 3074, 4068). Article estimates and sample sizes are from Table 7, columns 1–2.

Tell me how the course is going

- Anonymous, about 1 minute, all questions optional.
- Pick the week, your section, and rate the pace.
- Fill it in once for each week, if you like.

[Open the feedback form](#)



Scan with your phone camera

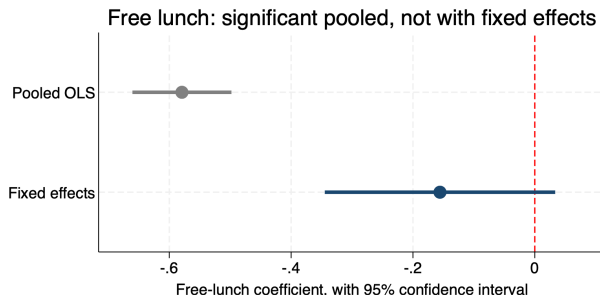
Appendix: why your Stata numbers are a little different

The class file `bcuse schoo193_98` is a close teaching version, not the exact estimation file used for the article.

- Our complete-case sample: 7,274 school-years in 1,773 schools.
- Papke's Table 7 sample: 7,242 school-years in 1,771 schools.
- Our coefficients: 7.64 (pooled) and 6.27 (fixed effects), vs. 8.442 and 7.179 in the article.

Focus first on the question each Stata command answers. The gap in exact estimates comes from the teaching data version, not a different FE idea.

Appendix: free lunch loses significance with fixed effects



Dot = estimate; bar = 95% confidence interval (the true values that fit our data).

- **Pooled:** -0.58 , interval $[-0.66, -0.50]$. Far from zero: ***.
- **Fixed effects:** -0.16 , interval $[-0.35, 0.03]$. It crosses zero, so we cannot rule out “no effect” ($p = 0.11$).

The bar is wider because poverty barely moves inside a school: less information.
Not significant $\neq$ poverty does not matter.

Appendix: clustered standard errors

`vce(cluster schid)` in both regressions. **Fixed effects** changes *which comparison* we make, so the coefficient changes. **Clustering** changes *how sure we are*, so only the standard errors change.

	Coef.	s.e. plain	s.e. clustered	<i>p</i> plain	<i>p</i> clustered
Spending, pooled	7.638	1.177	1.721	0.000	0.000
Spending, FE	6.270	2.092	2.432	0.003	0.010
Free lunch, pooled	-0.579	0.027	0.041	0.000	0.000
Free lunch, FE	-0.156	0.080	0.096	0.051	0.107

- Same coefficients. Standard errors 16–55% bigger.
- Why: 7,274 rows are really 1,773 schools seen 4–5 times. A school that beats our prediction one year tends to do so every year. Plain standard errors count those repeats as new information: overconfident.
- Free lunch (FE) would earn a star without clustering; honestly, it does not.

Appendix: why cluster, in plain words

The survey

Asking 20 different students gives 20 opinions. Asking the same 5 friends 4 times gives 20 answers, but not 20 independent opinions. Clustering counts friends, not answers.

Two rules

- **Cluster at the level that repeats:** here the school (schid), because each school appears in several years.
- **You need many clusters:** 1,773 schools is plenty. With 3 schools (Maple, Oak, Pine) clustered standard errors are unreliable, so the toy example does not cluster.

One line: clustering stops Stata from counting the same school as if it were several different schools.